Question 3 — Knowledge Distillation

1. Given problem

A teacher model and a student model predict the next word in:

> "The patient was admitted to the ___"

Index	Candidate	One-hot label
1	hospital	1
2	clinic	0
3	ward	0
4	school	0

The correct answer is hospital:

$$y = [1, 0, 0, 0]$$

Teacher logits:

$$z_t = [5.0, 3.0, 2.0, 0.0]$$

Student logits:

$$z_s = [3.0, 2.5, 1.0, 0.5]$$

Given: temperature $T = 2$, mixing weight $\alpha = 0.5$, learning rate $\eta = 0.5$, and natural logarithms throughout.

The assignment's exact loss is:

$$\boxed{L = \alpha T^2 KL(p_T \| q_T) + (1 - \alpha) CE(y, q_1)}$$

Here p_T is the teacher's softened distribution, q_T is the student's softened distribution, and q_1 is the student's ordinary probability distribution at $T = 1$.

Important: The KL term uses $T = 2$; hard-label cross-entropy uses q_1 at $T = 1$. Preserve the direction $KL(p_T \| q_T)$: teacher first, student second.

Part (a): Probabilities at T = 1

Use ordinary softmax:

$$P(i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

Teacher probabilities

Teacher logits:

$$z_t = [5, 3, 2, 0]$$

Exponentials:

Candidate	Exponential
Hospital	$e^5 = 148.413159102577$
Clinic	$e^3 = 20.085536923188$
Ward	$e^2 = 7.389056098931$
School	$e^0 = 1$

Denominator:

$$D_t = 148.413159102577 + 20.085536923188 + 7.389056098931 + 1$$

$$D_t = 176.887752124695$$

Divide each exponential by the denominator:

$$p_{1,1} = 148.413159102577 / 176.887752124695 = 0.839024507463$$

$$p_{1,2} = 20.085536923188 / 176.887752124695 = 0.113549619360$$

$$p_{1,3} = 7.389056098931 / 176.887752124695 = 0.041772570515$$

$$p_{1,4} = 1 / 176.887752124695 = 0.005653302662$$

$$\boxed{p_1 = [0.839024507463,\ 0.113549619360,\ 0.041772570515,\ 0.005653302662]}$$

Student probabilities

Student logits:

$$z_s = [3, 2.5, 1, 0.5]$$

Exponentials:

Candidate	Exponential
Hospital	$e^3 = 20.085536923188$
Clinic	$e^{2.5} = 12.182493960704$

Candidate	Exponential
Ward	$e^{1} = 2.718281828459$
School	$e^{0.5} = 1.648721270700$

Denominator:

$$D_s = 20.085536923188 + 12.182493960704 + 2.718281828459 + 1.648721270700$$

$$D_s = 36.635033983050$$

Probabilities:

$$q_{1,1} = 20.085536923188 / 36.635033983050 = 0.548260360083$$

$$q_{1,2} = 12.182493960704 / 36.635033983050 = 0.332536717895$$

$$q_{1,3} = 2.718281828459 / 36.635033983050 = 0.074198971119$$

$$q_{1,4} = 1.648721270700 / 36.635033983050 = 0.045003950903$$

$$\boxed{q_1 = [0.548260360083,\ 0.332536717895,\ 0.074198971119,\ 0.045003950903]}$$

Comparison

Candidate	Teacher T=1	Student T=1
Hospital	0.839024507463	0.548260360083
Clinic	0.113549619360	0.332536717895
Ward	0.041772570515	0.074198971119
School	0.005653302662	0.045003950903

Both select hospital, but the teacher assigns it substantially more probability. The student assigns more probability to the alternatives.

Part (b): Softened probabilities at T = 2

Use:

$$P_T(i) = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}}$$

Divide every logit by 2 before exponentiating.

Teacher at T = 2

$$z_t/T = [2.5, 1.5, 1, 0]$$

Exponentials:

$$[12.182493960704,\ 4.481689070338,\ 2.718281828459,\ 1]$$

Denominator:

$$D_{t,T=2} = 20.382464859501$$

Probabilities:

$$p_{T,1} = 12.182493960704/20.382464859501 = 0.597694834490$$

$$p_{T,2} = 4.481689070338/20.382464859501 = 0.219879641703$$

$$p_{T,3} = 2.718281828459/20.382464859501 = 0.133363744140$$

$$p_{T,4} = 1/20.382464859501 = 0.049061779667$$

$$\boxed{p_T = [0.597694834490,\ 0.219879641703,\ 0.133363744140,\ 0.049061779667]}$$

Student at T = 2

$$z_s/T = [1.5, 1.25, 0.5, 0.25]$$

Exponentials:

$$[4.481689070338,\ 3.490342957462,\ 1.648721270700,\ 1.284025416688]$$

Denominator:

$$D_{s,T=2} = 10.904778715188$$

Probabilities:

$$q_{T,1} = 4.481689070338/10.904778715188 = 0.410983953677$$

$$q_{T,2} = 3.490342957462/10.904778715188 = 0.320074624953$$

$$q_{T,3} = 1.648721270700/10.904778715188 = 0.151192547209$$

$$q_{T,4} = 1.284025416688/10.904778715188 = 0.117748874161$$

$$\boxed{q_T = [0.410983953677,\ 0.320074624953,\ 0.151192547209,\ 0.117748874161]}$$

Comparison of ordinary and softened distributions

Candidate	Teacher T=1	Teacher T=2	Student T=1	Student T=2
Hospital	0.839024507463	0.597694834490	0.548260360083	0.410983953677
Clinic	0.113549619360	0.219879641703	0.332536717895	0.320074624953
Ward	0.041772570515	0.133363744140	0.074198971119	0.151192547209
School	0.005653302662	0.049061779667	0.045003950903	0.117748874161

Temperature 2 softens both distributions. The ranking remains unchanged, but alternative candidates receive more probability.

Part (c): KL divergence

Use the exact direction specified by the assignment:

$$KL(p_T \| q_T) = \sum_i p_{T,i} \ln \left(\frac{p_{T,i}}{q_{T,i}} \right)$$

Teacher:

$$p_T = [0.597694834490,\ 0.219879641703,\ 0.133363744140,\ 0.049061779667]$$

Student:

$$q_T = [0.410983953677,\ 0.320074624953,\ 0.151192547209,\ 0.117748874161]$$

Individual contributions:

Candidate	KL contribution
Hospital	$0.597694834490 \ln(0.597694834490/0.410983953677) = 0.223852340393$
Clinic	$0.219879641703 \ln(0.219879641703/0.320074624953) = -0.082559057389$
Ward	$0.133363744140 \ln(0.133363744140/0.151192547209) = -0.016733663508$
School	$0.049061779667 \ln(0.049061779667/0.117748874161) = -0.042952305530$

Add:

$$\begin{aligned} KL(p_T \| q_T) = 0.223852340393 \\ - 0.082559057389 \\ - 0.016733663508 \\ - 0.042952305530 \end{aligned}$$

$$\boxed{KL(p_T \| q_T) = 0.081607313965}$$

The clinic, ward, and school contributions are negative because the student assigns those alternatives more probability than the teacher. Individual contributions can be negative; the complete KL divergence is non-negative.

Part (d): Hard-label cross-entropy

Use the ordinary student distribution q_1, not q_T:

$$CE(y, q_1) = -\sum_i y_i \ln q_{1,i}$$

Given:

$$y = [1, 0, 0, 0]$$

$$q_1 = [0.548260360083,\ 0.332536717895,\ 0.074198971119,\ 0.045003950903]$$

Substitute:

$$\begin{aligned} CE(y, q_1) = -[&1 \ln(0.548260360083) \\ &+ 0 \ln(0.332536717895) \\ &+ 0 \ln(0.074198971119) \\ &+ 0 \ln(0.045003950903)] \end{aligned}$$

The zero terms disappear:

$$CE(y, q_1) = -\ln(0.548260360083)$$

$$\boxed{CE(y, q_1) = 0.601004995223}$$

Only the hospital probability contributes because hospital is the correct class.

Part (e): Total distillation loss

The assignment gives:

$$L = \alpha T^2 KL(p_T \| q_T) + (1 - \alpha) CE(y, q_1)$$

Substitute:

$$L = 0.5(2^2)(0.081607313965) + (1 - 0.5)(0.601004995223)$$

Distillation contribution:

$$L_{KD} = 0.5 \times 4 \times 0.081607313965 = 0.163214627931$$

Hard-label contribution:

$$L_{CE} = 0.5 \times 0.601004995223 = 0.300502497612$$

Total:

$$L = 0.163214627931 + 0.300502497612$$

$$\boxed{L_{\text{initial}} = 0.463717125542}$$

Component	Value
Weighted distillation term	0.163214627931
Weighted hard-label term	0.300502497612
Initial total loss	**0.463717125542**

Part (f): Gradient with respect to student logits

We need the gradient of the **complete loss**, not only the KL component:

$$\nabla_{z_s} L$$

There are four student logits, so the gradient contains four components.

Softmax derivative

For ordinary softmax:

$$q_i = \frac{e^{z_i}}{\sum_k e^{z_k}}$$

The derivative is:

$$\boxed{\frac{\partial q_i}{\partial z_j} = q_i(\delta_{ij} - q_j)}$$

The Kronecker delta is 1 when $i = j$ and 0 otherwise. For temperature-scaled softmax, the chain rule adds a factor $1/T$:

$$\boxed{\frac{\partial q_{T,i}}{\partial z_{s,j}} = \frac{1}{T} q_{T,i}(\delta_{ij} - q_{T,j})}$$

Derive the KL gradient

Start with:

$$KL(p_T \| q_T) = \sum_i p_{T,i} \ln p_{T,i} - \sum_i p_{T,i} \ln q_{T,i}$$

The teacher is fixed, so the first term is constant with respect to student logits.

Differentiate the second term:

$$\frac{\partial KL}{\partial z_{s,j}} = -\sum_i \frac{p_{T,i}}{q_{T,i}} \frac{\partial q_{T,i}}{\partial z_{s,j}}$$

Substitute the temperature-softmax derivative:

$$= -\frac{1}{T} \sum_i p_{T,i} (\delta_{ij} - q_{T,j})$$

Because teacher probabilities sum to one:

$$\sum_i p_{T,i} = 1$$

The derivative becomes:

$$\boxed{\frac{\partial KL(p_T \| q_T)}{\partial z_{s,j}} = \frac{q_{T,j} - p_{T,j}}{T}}$$

The assignment multiplies this term by αT^2:

$$\frac{\partial}{\partial z_{s,j}} \left[\alpha T^2 KL \right] = \alpha T^2 \frac{q_{T,j} - p_{T,j}}{T}$$

Therefore:

$$\boxed{\nabla_{z_s} L_{KD} = \alpha T (q_T - p_T)}$$

Derive the cross-entropy gradient

Start with:

$$CE(y, q_1) = -\sum_i y_i \ln q_{1,i}$$

Use the ordinary softmax derivative:

$$\frac{\partial CE}{\partial z_{s,j}} = -\sum_i y_i(\delta_{ij} - q_{1,j})$$

Since the one-hot target sums to one:

$$\sum_i y_i = 1$$

We obtain:

$$\boxed{\nabla_{z_s} CE = q_1 - y}$$

Including the weight:

$$\boxed{\nabla_{z_s} L_{CE} = (1-\alpha)(q_1 - y)}$$

Combine both gradients

By the sum rule:

$$\boxed{\nabla_{z_s} L = \alpha T(q_T - p_T) + (1-\alpha)(q_1 - y)}$$

For $\alpha = 0.5$ and $T = 2$:

$$\boxed{\nabla_{z_s} L = (q_T - p_T) + 0.5(q_1 - y)}$$

Substitute the distributions:

$$p_T = [0.597694834490,\ 0.219879641703,\ 0.133363744140,\ 0.049061779667]$$

$$q_T = [0.410983953677,\ 0.320074624953,\ 0.151192547209,\ 0.117748874161]$$

$$q_1 = [0.548260360083,\ 0.332536717895,\ 0.074198971119,\ 0.045003950903]$$

$$y = [1, 0, 0, 0]$$

Hospital:

$$g_1 = (0.410983953677 - 0.597694834490) + 0.5(0.548260360083 - 1)$$

$$\boxed{g_1 = -0.412580700772}$$

Clinic:

$$g_2 = (0.320074624953 - 0.219879641703) + 0.5(0.332536717895)$$

$$\boxed{g_2 = 0.266463342198}$$

Ward:

$$g_3 = (0.151192547209 - 0.133363744140) + 0.5(0.074198971119)$$

$$\boxed{g_3 = 0.054928288629}$$

School:

$$g_4 = (0.117748874161 - 0.049061779667) + 0.5(0.045003950903)$$

$$\boxed{g_4 = 0.091189069946}$$

Complete gradient:

$$\boxed{\nabla_{z_s} L = [-0.412580700772,\ 0.266463342198,\ 0.054928288629,\ 0.091189069946]}$$

Candidate	Gradient	Effect under gradient descent
Hospital	-0.412580700772	Increase its logit
Clinic	+0.266463342198	Decrease its logit
Ward	+0.054928288629	Decrease its logit
School	+0.091189069946	Decrease its logit

A negative gradient means subtracting the gradient increases that logit; a positive gradient means subtracting it decreases that logit. The gradient entries sum to approximately zero because the probability distributions each sum to one.

Part (g): One gradient-descent step and new loss

Gradient descent updates the student logits:

$$z_{s,\text{new}} = z_{s,\text{old}} - \eta \nabla_{z_s} L$$

Given:

$$z_{s,\text{old}} = [3, 2.5, 1, 0.5], \qquad \eta = 0.5$$

Hospital:

$$z_{1,\text{new}} = 3 - 0.5(-0.412580700772) = 3.206290350386$$

Clinic:

$$z_{2,\text{new}} = 2.5 - 0.5(0.266463342198) = 2.366768328901$$

Ward:

$$z_{3,\text{new}} = 1 - 0.5(0.054928288629) = 0.972535855686$$

School:

$$z_{4,\text{new}} = 0.5 - 0.5(0.091189069946) = 0.454405465027$$

Updated student logits:

$$\boxed{z_{s,\text{new}} = [3.206290350386,\ 2.366768328901,\ 0.972535855686,\ 0.454405465027]}$$

New student probabilities at T = 1

Recalculate softmax from the updated logits:

$$q_{1,\text{new}} = softmax(z_{s,\text{new}})$$

$$\boxed{q_{1,\text{new}} = [0.623888748369,\ 0.269468107309,\ 0.066834376732,\ 0.039808767590]}$$

New softened student probabilities at T = 2

$$q_{T,\text{new}} = softmax(z_{s,\text{new}}/2)$$

$$\boxed{q_{T,\text{new}} = [0.447006174211,\ 0.293774185689,\ 0.146305245197,\ 0.112914394904]}$$

The teacher distribution remains unchanged:

$$p_T = [0.597694834490,\ 0.219879641703,\ 0.133363744140,\ 0.049061779667]$$

New KL divergence

$$KL_{\text{new}} = \sum_i p_{T,i} \ln\left(\frac{p_{T,i}}{q_{T,\text{new},i}}\right)$$

Candidate	Updated KL contribution
Hospital	0.173635075040
Clinic	-0.063705966687
Ward	-0.012351462657
School	-0.040895429314

$$\boxed{KL_{\text{new}} = 0.056682216382}$$

New hard-label cross-entropy

$$CE_{\text{new}} = -\ln(0.623888748369)$$

$$\boxed{CE_{\text{new}} = 0.471783214378}$$

New total loss

$$L_{\text{new}} = 0.5 \times 4 \times 0.056682216382 + 0.5 \times 0.471783214378$$

Distillation contribution:

$$0.5 \times 4 \times 0.056682216382 = 0.113364432765$$

Hard-label contribution:

$$0.5 \times 0.471783214378 = 0.235891607189$$

Total:

$$\boxed{L_{\text{new}} = 0.349256039954}$$

Before-and-after comparison

Quantity	Before update	After update
KL divergence	0.081607313965	0.056682216382
Cross-entropy	0.601004995223	0.471783214378
Total loss	0.463717125542	0.349256039954
Hospital probability at T=1	0.548260360083	0.623888748369

Absolute loss reduction:

$$0.463717125542 - 0.349256039954 = 0.114461085589$$

Percentage reduction:

$$\frac{0.463717125542 - 0.349256039954}{0.463717125542} \times 100$$

$$\boxed{\text{Loss reduction} \approx 24.6834\%}$$

The loss decreased after the update. The student's hospital probability increased, while both the teacher-to-student KL divergence and hard-label cross-entropy decreased. The update is applied to student logits, not directly to probabilities. Probabilities are recalculated from the updated logits. The teacher remains fixed.